

ONE WATER OPERATING SYSTEM | POWERED BY APAS.AI

One Water AI Executive Fellowship

Complete program book

24 weeks | 8 courses | 64 modules | Capstone defense

Fieldbook | Research workflow | Articulate and LearnWorlds delivery

Rebranded controlled document set | August 2026

A large, decorative graphic consisting of two overlapping circles in shades of blue, located in the bottom right corner of the page.

What is inside

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Naming correction

The current program is the One Water AI Executive Fellowship. The older 686-page One Water AI Master Class compilation remains a legacy source library. It is not relabeled as completed Fellowship instruction.

Professional judgment, not content access

The water sector does not need another general course that explains what artificial intelligence is. It needs leaders who can tell the difference between a demonstration and a defensible utility application. It needs operators, engineers, managers, data leaders, procurement teams, consultants, and executives who can work from the same language, test the same evidence, and identify the person who remains accountable.

The One Water AI Executive Fellowship is a six-month applied program for professionals responsible for water, wastewater, and stormwater decisions. Participants learn the technology in plain English, apply it to utility work, build with current tools, test failure and governance controls, and leave with an implementation portfolio for one real organizational opportunity.

This is not a collection of videos. It is a working program. Every course produces part of the participant's One Water AI Fieldbook. Every studio converts an idea into a decision, diagram, control, or work product. Every claim must carry its source and limitation. Every proposed use case must name its data, knowledge, risk, owner, measurement, stop condition, and next action.

What participants leave able to do

1. explain machine learning, generative AI, retrieval, knowledge graphs, skills, tools, agents, and agentic systems without hiding behind jargon;
2. distinguish chat, retrieval, deterministic workflow, single-agent, and multi-agent approaches;
3. diagnose whether a utility problem is ready for AI or still requires data, process, knowledge, governance, or organizational repair;
4. build and test prompts, reusable Claude Skills, source-grounded retrieval patterns, and bounded agent workflows;
5. evaluate output quality, provenance, permissions, failure modes, cost, and human-review effort;
6. identify valuable applications across operations, maintenance, water quality, customer service, capital delivery, regulation, finance, and administration;
7. create an AI governance and procurement record suitable for organizational review;
8. compare value, feasibility, consequence, adoption cost, and implementation risk;
9. lead cross-functional discussion between utility practice, data, technology, legal, security, procurement, and executive leadership; and
10. define and evaluate goals, environments, tools, memory, multi-agent handoffs, human authority, and stop conditions;
11. trace an AI application through development, release, monitoring, incident response, and retirement;
12. recognize synthetic or manipulated content and protect the provenance of public and internal records; and
13. defend a governed 90-day pilot and a one-year organizational roadmap.

Designed for consequential utility decisions

24 weeks	64 modules	132 hours	\$10,000 target
Six-month cohort	Eight courses	Guided effort	Per participant

Intended participants

- general managers, executive directors, and deputy directors;
- operations, maintenance, engineering, water-quality, stormwater, and wastewater leaders;
- capital-program, asset-management, customer-service, finance, and communications leaders;
- chief information, technology, data, cybersecurity, innovation, and transformation officers;
- legal, privacy, procurement, risk, audit, and governance professionals;
- senior consultants, solution providers, researchers, regulators, association leaders, and
- selected emerging leaders with the authority and organizational support to complete an applied

Current release boundary

This program book contains the complete rebranded curriculum and the production system that keeps it controlled. It does not claim that all 64 learner-facing Articulate modules are complete. Each module still passes research, factual, practitioner, learning, accessibility, and release gates.

Eight connected courses

The sequence moves from shared language to trusted context, bounded agents, utility applications, governance, economics, applied studios, and a defended pilot. Every module produces a Fieldbook record.

Course 1: AI Fluency for One Water

Build a shared language for AI and connect every term to a recognizable utility task, decision, or consequence.

Module	Learning job	Fieldbook result
1. AI and utility work	Separate useful assistance from accountable authority	AI use-case boundary card
2. How we arrived here	Understand rules, statistics, machine learning, deep learning, and generative AI as related but different approaches	AI family map
3. Models, tokens, prompts, and context	Explain what a model receives and produces without treating it as a person	prompt and context anatomy sheet
4. Language, images, audio, and multimodal AI	Recognize what changes when systems work across more than text	multimodal utility opportunity card
5. Chatbot, retrieval, workflow, agent, or agentic system	Choose the least complicated architecture that can perform a task	system classifier
6. Working effectively with an AI assistant	Frame a task, provide context, define output, test assumptions, and revise	reusable utility prompt pattern
7. Claude Skills and reusable organizational instructions	Understand a Skill as a versioned package of instructions, references, scripts, and assets that teaches a repeatable workflow	first utility Claude Skill specification
8. Fluency checkpoint	Explain a proposed AI use case to an operator, executive, technologist, and procurement professional	AI terms and decision field card

Course 2: Data, Knowledge, and Trusted Context

Understand why useful AI depends on connected records, shared meaning, current sources, permission, and traceable evidence.

Module	Learning job	Fieldbook result
9. Data before AI	Diagnose why AI amplifies disconnected or poor-quality data	readiness diagnosis
10. Data quality and observability	Identify completeness, accuracy, timeliness, consistency, and monitoring needs	data-quality control card
11. Metadata, lineage, and provenance	Explain where a record came from, how it changed, and why it may be trusted	lineage trace
12. Taxonomy and ontology	Build shared language and relationships without beginning with software	utility concept map
13. Knowledge graphs and the systems lens	See assets, people, records, decisions, sources, and consequences as connected relationships	knowledge relationship map
14. Embeddings and vector retrieval	Explain similarity search, what it can retrieve, and what it cannot prove	retrieval comparison sheet
15. Retrieval-augmented generation	Design a source-grounded answer path with citations and access boundaries	RAG answer contract
16. Context engineering and the trusted answer	Select the minimum useful context for a task and preserve its source and limitation	data and knowledge readiness map

Course 3: Agents, Skills, and Orchestration

See how AI moves from a response to bounded work, then design the instructions, tools, handoffs, permissions, and stop conditions required to keep people in control.

Module	Learning job	Fieldbook result
17. Before the agent	Compare a model response, chatbot, retrieval answer, workflow, and agent	architecture decision record
18. Inside the agent loop	Follow goal, environment, state, plan, action, observation, evaluation, adjustment, and stop	agent-loop trace
19. Agent anatomy	Examine model, instructions, environment, sources, tools, memory, identity, permissions, evaluation, and human owner	dependency and authority map
20. Tools, connectors, and the Model Context Protocol	Understand how an agent reaches approved systems and why connection does not equal authority	tool and connector register
21. Skills as organizational recipes	Design a reusable Skill with instructions, references, scripts, assets, triggers, and tests	working Claude Skill package
22. Memory, state, and durable records	Separate temporary context, task state, institutional memory, and authoritative records	state and record plan
23. Orchestration and multi-agent handoffs	Coordinate specialist roles, shared state, communication, disagreement, evidence, failure, and human escalation	handoff and conflict contract
24. Design the smallest useful agent	Choose agentiness according to value, consequence, readiness, review, interaction burden, and recovery needs	Utility Agent Canvas and agent experience map

Course 4: Utility Applications and Opportunity Design

Move from generic AI excitement to a disciplined portfolio of opportunities grounded in utility work, available evidence, affected people, and measurable value.

Module	Learning job	Fieldbook result
25. Operations and operator support	Examine bounded advisory, retrieval, shift, and exception-management patterns	operations opportunity card
26. Maintenance and asset management	Connect work orders, history, inspection, condition, failure, and planning	maintenance opportunity card
27. Water quality, laboratory, and compliance support	Trace evidence, limits, reviews, and certification boundaries	evidence-chain card
28. Customer service and public communication	Improve retrieval, consistency, routing, drafting, and accessibility while protecting privacy	customer-service use-case card
29. Capital planning, engineering, and project delivery	Connect requirements, risks, changes, schedules, records, and decision history	capital-delivery opportunity card
30. Stormwater, wastewater, drinking water, and One Water coordination	Examine cross-domain relationships without erasing domain-specific authority	One Water relationship map
31. Finance, procurement, legal, and administration	Identify high-volume knowledge work and its evidence and approval requirements	administrative opportunity card
32. Opportunity portfolio and prioritization	Compare value, feasibility, consequence, readiness, cost, and adoption effort	prioritized utility opportunity portfolio

Course 5: Governance, Security, and Human Authority

Build governance into the design so that the organization knows what the system may do, what it may not do, who checks it, and what evidence survives.

Module	Learning job	Fieldbook result
33. Risk and consequence tiers	Match review and control strength to possible harm and reversibility	use-case risk tier
34. Privacy, confidentiality, and data rights	Identify personal, sensitive, contractual, and restricted information boundaries	privacy and data-use record
35. Cybersecurity and adversarial failure	Recognize prompt injection, unsafe tool use, data leakage, poisoned context, and excessive privilege	threat and control map
36. Identity, permissions, and segregation of duties	Separate who asks, retrieves, recommends, approves, writes, and administers	permission matrix
37. Human authority, escalation, content integrity, and stop conditions	Name the accountable owner, recognize manipulated content, and design pause, refusal, provenance, and escalation rules	authority, content-integrity, and escalation map
38. Evaluation, testing, observability, and continuous assurance	Test usefulness, accuracy, source coverage, refusal behavior, resilience, cost, drift, traces, and incident signals	evaluation and observability plan
39. Vendor, model, and procurement governance	Compare claims, architecture, data use, service boundaries, exit terms, and evidence	AI procurement scorecard
40. The governed AI application record	Assemble purpose, sources, roles, controls, evaluations, release gates, monitoring, incidents, change history, retirement, and approvals	AI lifecycle governance record

Course 6: Strategy, Economics, and Organizational Change

Turn isolated experiments into a sequenced organizational program with value measures, ownership, funding, workforce preparation, and an operating rhythm.

Module	Learning job	Fieldbook result
41. Problem formulation before technology selection	Convert a complaint or idea into a measurable professional problem	problem statement
42. Value, baseline, and measurement	Define current effort, error, delay, risk, rework, service outcomes, benefit ownership, and measurement cadence	value baseline and benefits register
43. The full cost and operating stack of AI	Account for models, infrastructure, integration, data, LLMOps, observability, review, security, change, and failure	total-cost and operating-stack model
44. Build, buy, partner, or do nothing	Compare strategic control, speed, capability, dependency, and exit cost	sourcing decision record
45. AI operating model and decision rights	Define sponsors, product owners, technical owners, stewards, assurance, users, and escalation	operating-model canvas
46. Workforce capability and responsible adoption	Map changing tasks, participation, training, process redesign, communication, resistance, decision rights, and professional authority	adoption and workforce capability plan
47. Portfolio governance and sequencing	Fund learning in stages and stop weak initiatives before they become sunk costs	portfolio roadmap
48. Executive narrative and one-year roadmap	Explain the problem, opportunity, evidence, risk, investment, and sequence	investment case and one-year roadmap

Course 7: Applied AI Studios

Build enough to understand the mechanics, expose failure, and make better implementation decisions. No programming background is required.

Module	Learning job	Fieldbook result
49. Prompt and context studio	Improve an unclear utility task through role, context, constraints, output, and evaluation	tested prompt pattern
50. Source-grounded answer studio	Compare unsupported generation with governed retrieval and citations	RAG test record
51. Claude Skill studio	Build, trigger, test, revise, and document a reusable utility workflow	tested Claude Skill
52. Tool-use and permission studio	Connect a bounded read-only tool and inspect every request and result	tool-use evidence log
53. Agent workflow studio	Build a goal-to-result loop with a defined environment, tool policy, evaluation, retry limits, observability, and escalation	production-grade agent trace
54. Multi-agent decision-room studio	Observe specialist perspectives, shared evidence, disagreement, synthesis, and human decision	multi-agent decision record
55. Failure and red-team studio	Trigger unsupported claims, stale sources, instruction attacks, missing authority, and endless retries	failure and recovery log
56. Evaluation and pilot rehearsal	Test historical cases, record metrics, inspect human effort, exercise failure alerts and rollback, and prepare a controlled demonstration	release rehearsal evidence package

Course 8: Capstone, From Opportunity to Governed Pilot

Convert one real organizational opportunity into a pilot that leadership can understand, challenge, measure, govern, and stop when necessary.

Module	Learning job	Fieldbook result
57. Capstone charter and stakeholder agreement	Confirm the problem, sponsor, affected people, scope, and decision process	capstone charter
58. Current-state process and baseline	Document how the work happens today and where cost, delay, risk, or rework occurs	current-state map
59. Data and knowledge readiness	Identify approved sources, missing records, definitions, permissions, and provenance	readiness plan
60. Architecture and workflow decision	Select the least complicated technical pattern and justify every component	architecture decision record
61. Governance, security, and human authority	Complete the controls, permissions, review, incident, and escalation design	pilot control plan
62. Evaluation, economics, and success criteria	Define test cases, baseline, thresholds, cost limits, review effort, and stop conditions	measurement and financial plan
63. Ninety-day pilot and one-year adoption roadmap	Sequence discovery, build, test, review, decision, and possible expansion	90-day pilot brief and one-year roadmap
64. Capstone defense and professional commitment	Present the case, answer challenges, revise the plan, and state the next accountable action	defended implementation portfolio

Technical depth by readiness and role

The universal core remains 64 modules. A readiness diagnostic may route a participant to one or both optional 48-hour lines. These blueprints do not grant operating authority or authorize a credential.

One Water AI Builder Bridge

Give a motivated non-programmer or early builder enough technical understanding to read, change, test, connect, release, and defend a small governed utility application.

Module	Learning job	Applied result
BB01. Python and Computational Thinking for Utility Work	Read and explain a small program, break a utility task into clear steps, make a bounded change, run it, inspect the result, and correct an error without pretending code is self-explanatory.	Python task map and tested script evidence record
BB02. Data Structures, Files, and Application Data	Work with tables, lists, dictionaries, JavaScript Object Notation, files, dates, missing values, and identifiers while preserving the meaning, source, and limitation of utility records.	governed application-data contract and validation log
BB03. Application Programming Interfaces, Hypertext Transfer Protocol, and Tool Connections	Trace a governed request from an authenticated user through an application programming interface to an approved system, then inspect the response, error, permission, rate, and log.	governed tool-connection contract and request log
BB04. Git, GitHub, and Reproducible Work	Use version control to preserve who changed what, why it changed, how it was reviewed, what evidence passed, and how a weak change can be reversed.	reproducible change and review record
BB05. Testing, Dependencies, and Environments	Prove that a change works under known conditions, fails visibly under wrong conditions, and can be reproduced outside the builder's laptop.	test, dependency, and environment evidence package
BB06. Cloud Foundations, Identity, Secrets, and Cost	Explain and configure the smallest supportable hosted application while keeping identity, permission, data, secrets, logs, cost, resilience, and exit visible.	cloud responsibility, cost, and recovery map
BB07. Continuous Integration, Delivery, and Release Control	Move an approved change through automated checks, artifact creation, environment promotion, human approval, release, observation, and rollback.	continuous delivery and release-control record

Module	Learning job	Applied result
BB08. Builder Readiness Practicum	Bring code, data, tools, version control, tests, cloud responsibility, and release evidence together in a small read-only utility application that another person can reproduce and review.	reproducible read-only utility application and release evidence package

One Water AI Advanced Agent Systems

Give technical practitioners and reviewers a governed path through formal agent design, planning, retrieval, memory, multi-agent coordination, human interaction, learning methods, and embodied systems in water-sector settings.

Module	Learning job	Applied result
AA01. Agents, Goals, Environments, and Perceived Agency	Specify what an agent is trying to achieve, what it can observe and do, how the environment changes, how success is measured, and why people may perceive more agency than the system has.	advanced agent environment and authority model
AA02. Planning, Reasoning, Tool Use, and Stop Conditions	Compare planning patterns, inspect observable reasoning and action traces, control tool use, detect plan drift, and stop when evidence or authority fails.	advanced planning and stop-control specification
AA03. Agentic Retrieval and Evidence-Grounded Research	Build an iterative research and retrieval process that plans questions, selects approved sources, retrieves evidence, evaluates coverage, refines the search, and preserves conflicts and limitations.	agentic research plan, trace, claim register, and evidence package
AA04. Memory, State, Durable Records, and Correction	Separate temporary context, working state, episodic and semantic memory, learned parameters, and authoritative records so the system remembers only what it may keep and corrects history without erasing it.	agent state, memory, and authoritative-record plan
AA05. Multi-Agent Coordination, Conflict, and Failure	Design and test a multi-agent work system that preserves roles, evidence, communication, shared state, disagreement, arbitration, cost, failure recovery, and human command.	tested multi-agent coordination, evaluation, and recovery package
AA06. Human-Agent Interaction and Agentic User Experience	Design a usable and accessible interaction in which a person can understand status and evidence, control consequential action, challenge the system, interrupt it, correct it, recover, and retain authority.	accessible agentic user-experience and recovery specification
AA07. Reinforcement, Feedback, and Continual Learning	Explain how systems change from feedback, examples, rewards, fine-tuning, memory, retrieval, and software updates, then decide which forms of change are permitted and reviewed in a utility setting.	learning-method decision and safety record
AA08. Embodied Artificial Intelligence and Water Robotics	Evaluate how sensing, perception, language, planning, simulation, robotics, deterministic control, and human authority connect when an artificial intelligence system affects the physical world.	embodied-system evidence, control, and authority case

The capstone turns learning into a governed pilot

The capstone must identify and defend the following evidence.

1. the utility problem, current process, and baseline;
2. intended outcome, affected people, and measurable value;
3. accountable sponsor, owner, reviewers, users, and decision rights;
4. approved sources, data limitations, definitions, and provenance;
5. selected architecture and reason simpler alternatives were rejected;
6. tools, connections, identity, permissions, and records;
7. human checkpoints, prohibited actions, stop conditions, and escalation;
8. test cases, evaluation thresholds, incident response, and continuing assurance;
9. model, infrastructure, integration, review, workforce, and change costs;
10. the 90-day pilot sequence and one-year organizational roadmap; and
11. a candid statement of what remains unknown.

Program completion evidence

- completion of required Articulate modules and assessments;
- participation in at least six of eight live executive forums;
- completion of required studios and fieldwork;
- a completed One Water AI Fieldbook;
- a source-backed capstone portfolio;
- a capstone defense and revision; and
- an explicit statement of professional responsibility and next action.

Completion does not authorize engineering, operational, regulatory, financial, procurement, or security decisions. Any credential remains disabled until independent evidence, assessment, identity, and issuance gates are approved.

The Fieldbook keeps 64 modules connected

The One Water AI Fieldbook is part workbook, part laboratory notebook, part decision record, and part executive playbook. Every module adds one decision, map, test, reflection, or professional record. By Module 64, those records form the participant's implementation portfolio.

The repeating four-part rhythm

Before	What do I currently believe, and what decision would I make?
Learn	What relationship, mechanism, or distinction matters?
Apply	What does this mean for one utility task or organizational situation?
Keep	What record enters my implementation portfolio?

Participant control

Participants own their entries. Confidential organizational material remains private unless the participant explicitly shares it for review. Learning records can preserve completion events without ingesting confidential text.

From research question to governed lesson

1. Define value Name the utility decision, audience, work product, and professional value.	2. Discover sources Use Perplexity or another research tool to find source leads. Open the original sources.
3. Preserve evidence Register direct URLs, files, dates, permissions, and exact locators.	4. Verify claims Check every material claim against the original and record limits and conflicts.
5. Write the manuscript Produce one complete Markdown teaching paper in plain English.	6. Design graphics Describe visuals that explain processes, relationships, comparisons, and decisions.
7. Assemble learning Build in Articulate with interactions, explanatory feedback, and accessibility.	8. Review and release Pass factual, practitioner, learning, accessibility, and publishing gates.

Citation rule

Perplexity supports discovery. The original source supports the claim. Every material claim records the source title, issuer, date, direct URL or file, exact locator, support, limitation, currency, applicability, conflict, reviewer, and approval state.

Graphic rule

A graphic must make a relationship easier to understand. Each visual states what it shows, how the learner uses it, the conclusion the learner should reach, its accessible description, its mobile behavior, and the source IDs behind it. Decorative technology images do not count.

One source of truth, three delivery layers

	Original sources, transcripts, conversations, and practitioner judgment.
	Versioned syllabus, manuscript, claims, graphics brief, Fieldbook, and approvals.
	Learner instruction, interactions, quizzes, accessibility, and SCORM or xAPI package.
	Enrollment, cohort access, commerce, progress, community, and delivery.
	Governed concepts, relationships, sources, and deeper evidence exploration.

What changes together

When the curriculum changes, the syllabus, program book, curriculum prospectus, Fieldbook, metadata, and synchronization record are rebuilt together. Articulate and LearnWorlds receive only approved versions.

Release boundary and reference sources

The legacy boundary

The 686-page One Water AI Master Class compilation is valuable source material. Its old title, sequence, and module numbering do not match this Fellowship. Migration requires a module-by-module source map, claim review, voice review, visual review, and an explicit decision to retain, rewrite, combine, or retire each section.

Curriculum comparison sources

- MIT Sloan AI Executive Academy, current topics and sample schedule
<https://executive.mit.edu/ai-executive-academy.html>
- MIT Sloan AI Essentials, playbook, team simulation, action learning, governance, and adoption
<https://executive.mit.edu/ai-essentials.html>
- Anthropic, The Complete Guide to Building Skills for Claude
<https://resources.anthropic.com/hubfs/The-Complete-Guide-to-Building-Skill-for-Claude.pdf>

Document set

This package includes the self-contained HTML program book, this PDF program book, the 17-page curriculum prospectus, the 143-page Fieldbook working edition, the reusable module research and visual brief, and a package README.

One Water Operating System

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